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Methodological Issues Related to the Estimation of Financial Distress Prediction Models

MARK E. ZMIJEWSKI*

1. Introduction

Researchers typically estimate financial distress prediction models on nonrandom samples. Estimating models on such samples can result in biased parameter and probability estimates if appropriate estimation techniques are not used. This paper examines conceptually and empirically two estimation biases which can result when financial distress models are estimated on nonrandom samples. The first bias results from “oversampling” distressed firms and falls within the topic of *choice-based sample biases*. The second results from using a “complete data” sample selection criterion and falls within the topic of *sample selection biases*.

The two issues examined in this paper arise because of sample selection/data collection constraints typically faced by financial distress researchers. The first constraint is the extremely low frequency rate of firms exhibiting financial distress characteristics (e.g., petitioning for bankruptcy) in the population.¹ The second is that data for financially distressed firms are often unavailable. A summary of the sample selection/data collection procedures used in 17 financial distress studies is

* University of Chicago. The topics addressed in this paper were suggested by John Abowd. I am indebted to Robert Holthausen, Richard Leftwich, and Katherine Schipper for their guidance and suggestions which were used to refocus this paper, and to Craig Olson for providing the computer algorithms used in the analyses. I would also like to thank Lawrence Brown, George Foster, Robert Hagerman, Philip Perry, and Charles Trzcinka for their comments on an earlier draft.

¹ Dun and Bradstreet report that the frequency rate of business failures in the United States has never exceeded .75% since 1934 (*Business Failure Record* [1982]).

presented in table 1 to illustrate how samples have been collected in this literature.² With regard to sample selection, all 17 studies estimated models on the "known" population of distressed firms subject to a complete data criterion, with 12 of the 17 studies using a matched-pairs design for collecting the nondistressed sample. With regard to sample composition, an examination of the distressed firm sample frequency rates indicates that only three studies use rates of less than 40%, with 11 studies using a 50% rate. Thus, it is clear that these studies estimated models on nonrandom samples which have compositions considerably different from the population's composition.

Section 3 of the paper examines the effect of choice-based sample selection on model estimation. Most estimation techniques assume the use of exogenous random sampling designs in which an observation is randomly drawn and the dependent and independent variables are observed. In contrast, a choice-based sample results when the probability of an observation entering the sample depends on the value (attributes or group) of the dependent variable. That is, a researcher first observes the dependent variable and then draws the sample based on that knowledge. This approach violates the random sampling design assumption and causes both parameter and probability estimates to be asymptotically biased (see Manski and Lerman [1977] and Palepu [1983]). The observed result of this bias is that a dependent variable group having a sample probability larger than the population probability is oversampled, with the oversampled group having understated classification and prediction error rates. (The converse statement holds for a dependent variable group having a sample probability smaller than the population probability.)

Recall that the population frequency of financial failure has not exceeded .75% since 1934 (see n. 1). Thus, all studies reviewed in table 1 use overstated distressed firm sample frequency rates. The rates range from 1.5% (two times the population rate) to 50.0% (66.7 times the population rate). Pearson correlation coefficients between the sample frequency rate and the group error rates reported in table 1 are used to indicate the existence of a choice-based sample bias. Correlation coefficients consistent with this bias would be negative for distressed firms (i.e., higher distressed firm sample frequency rates cause lower distressed firm estimated error rates) and positive for nondistressed firms (i.e., higher distressed firm sample frequency rates cause higher nondistressed firm estimated error rates). A zero correlation indicates no association

² Studies conducted within specific industries or on special types of firms have not been summarized in table 1. Examples of excluded studies are Altman [1973] and White and Turnbull [1976], who have conducted research within the railroad industry; Edmister and Schlarbaum [1974], Sinkey [1975; 1977], Martin [1977], Santomero and Vinso [1977], and Pettway and Sinkey [1980], within the banking industry; Altman and Lorris [1975], within the over-the-counter broker-dealer industry; Edmister [1972], on small business failures, Schipper [1977] and Shrieves and Stevens [1979], within the education industry; and Pinches and Trieschmann [1974], within the insurance industry.

TABLE 1
Summary of Sample Selection/Data Collection Procedures Used in Financial Distress Research

Research Study	Sample Selection			Sample Composition			Classification Error Rates ⁴	
	Complete Data Criterion	Population of Distressed Firms Used	Matched Pairs Sample	Number of Firms		Sample Frequency Rate ³	D ¹	ND ²
				D ¹	ND ²			
Beaver [1966]	Yes	Yes	Yes	79	79	.500	NR	NR
Altman [1968]	Yes	Yes	Yes	33	33	.500	.06	.03
Wilcox [1971;1973]	Yes	Yes	Yes	52	52	.500	NR	NR
Deakin [1972]	Yes	Yes	Yes	32	32	.500	.03	.03
Blum [1974]	Yes	Yes	Yes	115	115	.500	NR	NR
Elam [1975]	Yes	Yes	Yes	48	48	.500	NR	NR
White and Turnbull [1975]	Yes	Yes	No	34	2,303	.015	.53	.01
Altman et al. [1977]	Yes	Yes	Yes	53	58	.477	.06	.09
Deakin [1977]	Yes	Yes	No	63	80	.441	.12	.01
Ketz [1978]	Yes	Yes	No	75	100	.429	.33	.04
Van-Frederikslust [1978]	Yes	Yes	Yes	20	20	.500	.05	.10
Norton and Smith [1979]	Yes	Yes	Yes	30	30	.500	.11	.03
Dambolena and Khoury [1982]	Yes	Yes	No	46	46	.500	.09	.01
Ohlson [1980]	Yes	Yes	Yes	105	2,058	.049	NR	NR
Emery and Cogger [1982]	Yes	Yes	Yes	52	52	.500	NR	NR
Zavgren [1982]	Yes	Yes	Yes	45	45	.500	NR	NR
Zmijewski [1983]	Yes	Yes	No	96	3,880	.025	.83	.01

¹ D-distressed firms.² ND—nondistressed firms.³ Number of distressed firms/total number of firms in sample. The Pearson correlation coefficient between the distressed firm sample frequency rate and the distressed and nondistressed classification error rates are -.932 and .428, significant at the .01 and > .10 levels, respectively.⁴ Error rates are the proportion of firms incorrectly classified reported in the respective studies for one year prior to financial distress.

between sample frequency rates and group error rates and is consistent with unbiased estimates. The correlations reported in table 1 indicate the existence of a bias: the correlation between the sample frequency rate and the distressed firm error rate is negative ($-.932$) and significant at the .01 level. The correlation between the sample frequency rate and the nondistressed firm error rate is positive ($.428$), although not significant at the .10 level.³ These results indicate that group error rates are associated with sample frequency rates and provide at least a partial explanation for the divergent distressed firm error rates reported in previous financial distress studies.

This comment is not meant to imply that choice-based sample designs are necessarily inappropriate. Samples collected in this manner are often cost effective when certain phenomena occur infrequently and there is a high cost in collecting a random sample containing a sufficient number of "rare event" observations to estimate the model. However, these samples violate the exogenous random sample assumption of most estimation techniques, and alternative or adjusted techniques are needed to estimate unbiased parameters. Although using an appropriate estimation technique may not improve overall classification and prediction error rates, it will provide unbiased parameter and probability estimates allowing users both to assess the effect of individual variables and to choose optimal cutoff probabilities based on individual loss functions.

Section 4 examines the effect of the sample selection bias resulting from the use of a complete data sample selection criterion. Estimating models on such samples using techniques which assume random sampling can also result in an estimation bias. If incomplete data observations are nonrandomly distributed in the population and this information is not incorporated into estimating the model, both the estimated parameters and the probabilities will be asymptotically biased (see Heckman [1979], Poirier [1980], and Maddala [1983]). The bias is a function of the nature of the nonrandomness of the data. For example, suppose distressed firms are more likely to have incomplete data, that is, the probability of distress and the probability of complete data have a negative covariance. Estimating a model conditional on complete data and ignoring this information results in estimated probabilities which, on average, understate the population probabilities. The probability of distress given complete data is less than the probability of distress of an observation randomly drawn from the population, because observations with high probabilities of distress have, on average, low probabilities of being selected for the sample (i.e., of having complete data).

The above supposition is not unreasonable. Dun and Bradstreet report

³ This test assumes that the divergent error rates are not due to sample differences. Given that the 17 studies all use the population of distressed firms and have overlapping periods of study, this assumption appears reasonable.

that financially failed firms tend to be both younger and smaller.⁴ Thus it is reasonable that such firms are also less likely to have complete data. Unfortunately, no previous financial distress study (of which I am aware) examines the problem of incomplete data. However, some indirect inferences can be drawn from previous studies examining more than one forecast horizon. The distressed firm sample attrition rates for those studies were as high as 53% over an eight-year forecast horizon (Blum [1974]). This attrition indicates the potential for a missing data problem for financially distressed firms.

The empirical tests conducted in this paper to examine these two issues provide the following conclusions. With regard to the choice-based sample bias issue, the results clearly demonstrate the existence of a bias, a decrease in the bias as the sample selection probability approaches the population probability, and the elimination of the bias if appropriate estimation techniques are used. With regard to the sample selection bias issue, the results indicate the existence of a significant sample selection bias in the majority of the tests conducted. However, for both issues, the results do not indicate significant changes in overall classification and prediction rates, nor do they indicate different qualitative results (statistical inferences) for the financial distress model tested. Only the individual group classification and prediction rates were significantly affected.

Section 2 describes the sample selection procedures, data sources, and the data used in the empirical analyses. A more formal discussion of the choice-based sample and sample selection biases, as well as a discussion of the results of empirical tests attempting to ascertain the effects of these biases, is presented in sections 3 and 4, respectively. A summary and concluding comments are presented in section 5.

2. Sample Selection and Data Sources

Empirical tests are conducted on a financial distress model in sections 3 and 4 which require samples of financially distressed and nondistressed firms. Certain population characteristics (e.g., financial distress frequency rates) are also required. Before data can be collected, the population of firms must be identified and an operational definition of financial distress assumed. The population of firms for this study consists of all firms listed on the American and New York Stock Exchanges during the period 1972 through 1978 which have industry (SIC) codes of less than 6000.⁵ The number of firms in the population ranges from 2,082 to 2,241 per year. Financial distress is defined as the act of filing a

⁴ *Business Failure Record* [1982].

⁵ This period of study was chosen for the following reason. The SEC 10K report collection available at Cornell University is the data source for financially distressed firms. Although this collection has been available since 1971, the 1971 collection is incomplete and 1972 is the first year for which data are collected; 1978 is the last year data were available at the time these data were collected.

petition for bankruptcy. A firm is identified as bankrupt if it filed a bankruptcy petition during this period and nonbankrupt if it did not.

Each firm in the population was identified as either bankrupt or nonbankrupt using three sources: the *Capital Changes Reporter*, the *Wall Street Journal Index*, and the *Compustat Research File*.⁶ The total number of firms identified as bankrupt was 129, of which 81 satisfied the complete data criterion for estimating the financial distress model. The distributions of these samples across years are reported in table 2. The distributions for the total and complete data samples are similar, indicating that bankrupt firms with incomplete data occur randomly across years.⁷ Both samples also have distributions similar to the business failure distribution reported by Dun and Bradstreet for this period.⁸ The bankruptcy frequency rates in the population range from .49% to .94%.

Two types of data are required for estimating the models examined in this paper. The first type consists of one variable, the period of listing on the *CRSP Daily Return File*, collected, obviously, from that source. This variable is available for the entire population of firms. The second type of data is financial statement data. For the bankrupt sample, these data were collected from the last SEC 10K report filed before the date a firm petitioned for bankruptcy. Data were available for approximately 63% of the bankrupt sample. For the nonbankrupt sample, these data were collected from the *Compustat Annual Industrial File*, with approximately 67% of the firms having complete data. Data for approximately 33% of the nonbankrupt firms are not available because *Compustat* includes only firms which meet certain size criteria; hence small firms are not listed. Data for the nonbankrupt sample were collected for each year (1972 through 1978).

TABLE 2
Distribution of Bankrupt Firms by Year of Bankruptcy Petition

Year	Total Sample Identified		Sample with Complete Data		Sample with Incomplete Data	
	Number	Percentage	Number	Percentage	Number	Percentage
1972	10	7.8	7	8.6	3	6.3
1973	16	12.4	10	12.3	6	12.5
1974	18	13.9	13	16.1	5	10.4
1975	17	13.2	11	13.6	6	12.5
1976	21	16.3	13	16.1	8	16.7
1977	20	15.5	10	12.3	10	20.8
1978	27	20.9	17	21.0	10	20.8
Total	129	100.0	81	100.0	48	100.0

⁶ It is assumed that these sources identify all bankrupt firms in the population. To the extent they do not correctly identify bankrupt firms, the results may be biased.

⁷ The chi-square test statistic comparing the total sample with the complete data sample is .55, which is not significant at the .90 level.

⁸ *Business Failure Record* [1982].

3. Nonrandom Sampling: Choice-Based Sample Bias

The sampling procedure used most often in financial distress studies is first to identify, as distinct populations (groups), distressed and non-distressed firms, and then to select firms from within each population separately. This choice-based sampling technique results in nonrandom samples because the probability of a firm entering the sample depends on the observed financial distress attributes of the firm. Under the strict assumptions of discriminant analysis, only the constant term is affected by estimating models on such samples, and an adjustment can be made to the classification rule by incorporating the population frequency rate of financial distress into the rule (see Morrison [1976, chap. 6]).⁹

However, for most specifications of selection probability type models (i.e., logit and probit model specifications), the constant and all of the coefficients are asymptotically biased. The inconsistent and biased estimates which result from using unadjusted probit or logit assessments on choice-based samples are demonstrated in Manski and Lerman [1977], Cosslett [1981], Manski and McFadden [1981], and Palepu [1983]. There are primarily three available estimation techniques which are appropriate for estimating models using choice-based samples: weighted exogenous sample maximum likelihood (*WESML*), conditional maximum likelihood (*CML*), and full information concentrated maximum likelihood (*FICML*). All three techniques provide asymptotically consistent normal parameter estimates; however, only *FICML*'s estimates are asymptotically efficient.¹⁰ Computationally, *WESML* is the least complex of these techniques and is used to examine this issue. The financial distress model used in this study and the *WESML* technique are briefly outlined below.¹¹

Assume that equation (1) appropriately describes the probability of bankruptcy as a probit equation where a firm is observed to file a petition for bankruptcy ($B = 1$) when B^* , an underlying response variable, exceeds zero:

$$P(B = 1) = P(B^* > 0) \quad (1a)$$

$$B^* = a_0 + a_1ROA + a_2FINL + a_3LIQ + u \quad (1b)$$

$$P(B^* > 0) = P(-u < a_0 + a_1ROA + a_2FINL + a_3LIQ) \quad (1c)$$

where:

$P(\cdot)$ = probability of $(\cdot)$,

⁹ Altman, Haldeman, and Narayanan [1977] made such adjustments when classifying firms.

¹⁰ The statistical properties for these techniques are derived in Hsieh, Manski, and McFadden [1983] and are compared empirically in McFadden, Hsieh, and Manski [1983]. The empirical comparison indicates virtually identical results for *CML* and *FICML*, with some variation in the *WESML* results. However, the *WESML* results are qualitatively identical to the *CML* and *FICML* results.

¹¹ This technique is more fully developed in Manski and Lerman [1977].

$B = 1$ if bankrupt, 0 otherwise,
 ROA = net income to total assets (return on assets),
 $FINL$ = total debt to total assets (financial leverage),
 LIQ = current assets to current liabilities (liquidity), and
 u = a normally distributed error term.¹²

It is well known that under the assumption of random exogenous sampling, probit assessment parameters are estimated for equation (1) by maximizing the log-likelihood function (L^*) shown in equation (2):

$$L^* = \sum_j (B) \ln[\Phi(H)] + \sum_j (1 - B) \ln[1 - \Phi(H)] \quad (2)$$

where:

$$\begin{aligned} \Phi &= \text{cumulative density function for a standard} \\ &\quad \text{normal variable,} \\ H &= a_0 + a_1 ROA + a_2 FINL + a_3 LIQ, \end{aligned}$$

and all other variables are as defined in equation (1). However, this approach assumes that the sample and population compositions are identical. In contrast to this approach, the *WESML* probit assessment weights the log-likelihood function by the ratio of the population frequency rate to the sample frequency rate of the individual groups. Weighting the log-likelihood function adjusts the parameters for the choice-based sample. The weighted log-likelihood function which is maximized in the *WESML* probit assessment is shown in equation (3):

$$\begin{aligned} L^* &= [POP/SAMP] \sum_j (B) \ln[\Phi(H)] \\ &\quad + [(1 - POP)/(1 - SAMP)] \sum_j (1 - B) \ln[1 - \Phi(H)] \end{aligned} \quad (3)$$

where:

$$\begin{aligned} POP &= \text{the proportion of bankrupt firms in the population,} \\ SAMP &= \text{the proportion of bankrupt firms in the sample,} \end{aligned}$$

and all other variables are as defined above.

It is clear from equation (3) that as the sample selection probability approaches the population probability, the likelihood function using *WESML* approaches the “unweighted” probit likelihood function (i.e., $POP/SAMP$ and $(1 - POP)/(1 - SAMP)$ both approach one). Thus, studies using sample selection probabilities close to the population probability, such as White and Turnbull [1975], Ohlson [1980], and Zmijewski

¹² This paper does not address the financial distress model formulation criticisms made previously in the literature. The focus is on the appropriate estimation of such models. Models that appear in this paper are developed for demonstrative purposes only and are not an attempt to develop premier prediction models. The variables were chosen because of their performance, reported in previous research studies.

[1983], are not as affected by this problem as studies using 50% sample frequency rates (see table 1). Studies using a 50% sample frequency rate oversample bankrupt firms and have sample selection probabilities much different from the population probability (i.e., $POP/SAMP \ll 1$ and $(1 - POP)/(1 - SAMP) \gg 1$). The result is upwardly biased probability estimates and, thus, an overclassification of bankrupt firms.

The choice-based sampling issue is examined empirically by estimating the model shown in equation (1) using both the *WESML* and the unweighted probit assessments on six choice-based estimation samples. The samples have decreasing bankrupt firm frequency rates, so the bias induced by estimating the model via an unadjusted probit assessment can be examined as the sample selection probabilities approach the population probability. The population of firms, data sources, and the data collection procedures are as described in section 2. The bankrupt sample with complete data (81 firms) was pooled across years. For consistency, a year (1972 through 1978) was assigned randomly to each of the nonbankrupt firms with complete data, and the nonbankrupt firms were also pooled across years.¹³ This resulted in a nonbankrupt firm sample size of 1,600 and a total sample of 1,681 firms.¹⁴ Since all firms in the population were identified as either bankrupt or nonbankrupt (see section 2), the population frequency rate of bankruptcy over this period is known; this rate is equal to .847%.¹⁵

The first step in selecting estimation samples is to randomly partition the total sample into two subsamples, an "estimation sample" containing 40 bankrupt and 800 nonbankrupt firms, and a "prediction sample" containing 41 bankrupt and 800 nonbankrupt firms. The estimation sample is used to select the alternative choice-based estimation subsamples; the prediction sample is used to compare the *WESML* and unweighted probit assessments on the basis of predicted probabilities. Each of the choice-based samples has different compositions (i.e., bankrupt firm frequency rates), making classification comparisons across samples difficult; however, the prediction sample is identical across all estimations and is a good benchmark for comparison.

Six choice-based samples were drawn from the estimation sample described above. The six samples each contain all 40 bankrupt firms but different (increasing) numbers of nonbankrupt firms, selected randomly

¹³ Pooling firms across years assumes that the model is stationary, which is most likely violated (see Zmijewski [1983]). However, a violation of this assumption should not affect the issue under study since both estimation techniques are affected similarly.

¹⁴ The effects of incomplete data were not addressed in this section. Thus, the model estimated in this section is actually the model conditional on complete data and not the population model. Although this would cause a problem if one were attempting to make inferences regarding the population, it should not affect the issue specifically addressed here. I am aware of no estimation technique which addresses these issues simultaneously.

¹⁵ If the population frequency rate is unknown, techniques are available to estimate this parameter; see Hsieh, Manski, and McFadden [1983].

from the 800 available nonbankrupt firms in the estimation sample. The number of nonbankrupt firms in the six estimation samples is 40, 100, 200, 400, 600, and 800. The resulting bankrupt firm frequency rates are .500 (largest bias), .286, .167, .091, .063, and .048 (smallest bias), respectively.

The existence of a choice-based sample bias from using unweighted probit and the effects of the *WESML* adjustment are examined by comparing the estimated parameters, the correct classification rates, and the correct prediction rates across estimation samples and techniques. The equation (1) estimation results are reported in table 3, the classifications and classification comparisons in table 4, and the predictions and prediction comparisons in table 5. The results reported in these tables indicate the existence of a choice-based sample bias from using unweighted probit and the elimination of the bias when using *WESML* probit. The results also show a reduction in the bias using unweighted probit as the sample selection probability approaches the population probability.

The statistical inferences for the overall model and individual coefficients are similar for both the unweighted and *WESML* assessments across all six estimation samples (see table 3). Such similarities between the two techniques may be due to either the nature of the empirical associations examined in the model, or they may be due to the similarity between the estimation techniques themselves. Thus, it is unclear under what conditions *WESML* assessments would provide different statistical inferences from those provided using unweighted probit.

The existence of a bias is examined by comparing the unweighted probit results across the six estimation samples. If a bias does not exist, then there should be no relation between the bankrupt firm sample frequency rate of the choice-based estimation samples and the various results from the unweighted probit assessment (e.g., estimated parameters, group classification and prediction error rates, etc.). Evidence indicating a functional relationship between the sample frequency rate and the various probit results is consistent with a bias. The direction of the relationship if a bias exists can be specified for the estimated constant and the individual group classification and prediction rates. On average, lower bankrupt firm sample frequency rates result in lower probability estimates. Therefore, lower bankrupt firm sample frequency rates should result in lower estimated constants (positive association), lower bankrupt firm correct classification and prediction rates (positive associations), and higher bankrupt firm correct classification and prediction rates (negative associations). Pearson correlation coefficients between the choice-based sample frequency rate and the various probit results are used to indicate the existence of a functional relationship and, hence, a bias.

TABLE 3
Comparison of Unweighted and WESML Probit Model Estimations Across Alternative Estimation Samples

	CHOICE-BASED SAMPLES ¹					Pearson Correlation Coefficient ²
	40:40 (.500)	40:100 (.286)	40:200 (.167)	40:400 (.691)	40:600 (.063)	40:800 (.048)
<i>Panel A—Unweighted Probit</i>						
COEFFICIENTS ³						
CONSTANT	−2.953 ^a	−3.674	−4.206 ^a	−4.288 ^a	−5.503 ^a	−4.336 ^a
ROA	−9.023 ^a	−5.673 ^a	−6.669 ^a	−3.720 ^a	−3.918 ^a	−4.513 ^a
FINL	5.503 ^a	5.606 ^a	6.083 ^a	6.126 ^a	5.917 ^a	5.679 ^a
LIQ	.197	.186	.219	.052	.047	.004
−2xLog-Likelihood Ratio ⁴	76.89 ^a	107.74 ^a	147.35 ^a	176.24 ^a	190.87 ^a	203.78 ^a
<i>Panel B—WESML Probit</i>						
COEFFICIENTS ³						
CONSTANT	−4.777 ^a	−5.789 ^a	−5.922 ^a	−5.581 ^a	−5.106 ^a	−4.803 ^a
ROA	−8.702 ^a	−4.238 ^a	−5.019 ^a	−2.479 ^a	−2.739 ^a	−3.599 ^a
FINL	5.133 ^a	6.236 ^a	6.574 ^a	6.496 ^a	5.725 ^a	5.406 ^a
LIQ	−.002	.146	.213	.031	−.061	−.100
−2xLog-Likelihood Ratio ⁴	107.99 ^a	161.61 ^a	207.30 ^a	250.30 ^a	272.15 ^a	286.42 ^a

¹ Number of bankrupt: number of nonbankrupt firms in the choice-based estimation sample. The bankrupt firm sample frequency rate (number of bankrupt firms/total number of sample firms) is reported in parentheses.

² Pearson correlation coefficients between the estimation sample frequency rate and the coefficient reported in the corresponding row. NA indicates that the correlation test is not applicable.

³ Estimated probit parameters for equation (1): ROA = net income to total assets (return on assets); FINL = total debt to total assets (financial leverage); and LIQ = current assets to current liabilities (liquidity).

⁴ Goodness-of-fit test distributed as a chi-square with three degrees of freedom.

^a Significant at the .01 level.

^b Significant at the .05 level.

^c Significant at the .10 level.

TABLE 4
Comparison of Unweighted and WESML Probit Model Classifications Across Alternative Estimation Samples¹

	CHOICE-BASED SAMPLES ²					Pearson Correlation Coefficient ³
	40:40 (.500)	40:100 (.206)	40:200 (.167)	40:400 (.091)	40:600 (.063)	40:800 (.048)
<i>Panel A—Unweighted Probit</i>						
CLASSIFICATIONS ⁴						
BANKRUPT	92.5	77.5	77.5	72.5	65.0	62.5
NONBANKRUPT	100.0	100.0	99.0	99.3	99.5	99.5
OVERALL	96.3	93.6	95.4	96.8	97.3	97.7
CHI-SQUARE ⁵	65.17 ^a	95.10 ^a	158.10 ^a	267.05 ^a	345.88 ^a	420.93 ^a
						.949 ^a
						.681 ^c
						NA
						NA
<i>Panel B—WESML Probit</i>						
CLASSIFICATIONS ⁴						
BANKRUPT	52.5	47.5	50.0	48.0	42.5	42.5
NONBANKRUPT	100.0	100.0	99.5	100.0	99.8	99.9
OVERALL	76.3	85.0	91.3	95.0	96.3	97.1
CHI-SQUARE ⁵	25.83 ^a	50.98 ^a	96.19 ^a	176.38 ^a	230.62 ^a	306.32 ^a
						NA
<i>Panel C—Comparisons</i>						
PERCENT THE SAME ⁶	80.0	91.4	95.0	96.8	98.3	98.7
CHI-SQUARE ⁷	30.23 ^a	121.6 ^a	136.49 ^a	225.17 ^a	367.84 ^a	485.23 ^a
						−.989 ^a
						−.833 ^a

¹ Classifications are based on the estimated probabilities of each estimation sample for a given technique using a .5 probability cutoff, that is, firms with probabilities greater than or equal to (less than) .5 are classified as bankrupt (nonbankrupt).
² Number of bankrupt: number of nonbankrupt firms in the choice-based estimation sample. The bankrupt firm sample frequency rate (number of bankrupt firms/total number of sample firms) is reported in parentheses.
³ Pearson correlation coefficients between the estimation sample frequency rate and the result reported in the corresponding row. NA indicates that the correlation test is not applicable.

⁴ Percentage of firms correctly classified.

⁵ Chi-square test comparing actual with classified states.

⁶ The percentage of firms classified to be in the same state by both the WESML and unadjusted probit assessments.

⁷ Chi-square test comparing the WESML probit classifications with the unadjusted probit classifications.

^a Significant at the .01 level.

^b Significant at the .05 level.

^c Significant at the .10 level.

TABLE 5
Comparison of Unweighted and WESML Probit Predictions Across Alternative Estimation Samples¹

	CHOICE-BASED ESTIMATION SAMPLE ³						Pearson Correlation Coefficient ²
	40:40 (.500)	40:100 (.286)	40:200 (.167)	40:400 (.091)	40:600 (.063)	40:800 (.048)	
<i>Panel A—Unweighted Probit</i>							
PREDICTIONS ⁴							
BANKRUPT	97.6	90.2	90.2	78.0	70.7	70.7	.895 ^a
NONBANKRUPT	92.5	97.9	98.6	99.0	99.1	99.5	-.947 ^a
OVERALL	92.6	97.6	98.3	98.1	97.9	98.2	-.892 ^a
CHI-SQUARE ⁵	293.41 ^a	489.45 ^a	555.95 ^a	494.24 ^a	447.61 ^a	491.82 ^a	-.735 ^b
AVE. PROBABILITY ⁶	18.7	11.1	8.9	6.8	5.6	4.9	.995 ^a
GOODNESS OF FIT ⁷	1869.38 ^a	355.69 ^a	133.41 ^a	38.05 ^a	11.61	5.74	.940 ^a
<i>Panel B—WESML Probit</i>							
PREDICTIONS ⁴							
BANKRUPT	.538	48.8	51.2	46.3	41.5	43.9	46.3
NONBANKRUPT	99.8	99.6	99.6	99.6	100.0	100.0	-.281
OVERALL	97.3	97.1	97.4	97.1	97.3	97.4	-.129
CHI-SQUARE ⁵	375.58 ^a	305.69 ^a	345.58 ^a	305.69 ^a	317.94 ^a	338.25 ^a	.598
AVE. PROBABILITY ⁶	3.2	2.9	3.3	3.0	2.6	2.6	.601
GOODNESS OF FIT	9.86	13.17 ^b	9.48	9.89	12.01	11.94	-.199
<i>Panel C—Comparisons</i>							
PERCENT THE SAME ⁸	90.9	96.3	97.3	97.7	97.9	98.3	-.959 ^a
CHI-SQUARE ⁹	166.68 ^a	313.44 ^a	390.3 ^a	389.94 ^a	364.50 ^a	424.70 ^a	-.954 ^a

¹ Predictions are based on estimated probabilities of the prediction sample (a holdout sample of 41 bankrupt and 800 nonbankrupt firms) using parameters estimated on the respective estimation technique/choice-based sample combination. Firms with probabilities greater than or equal to (less than) .5 are predicted as bankrupt (nonbankrupt).
² Pearson correlation coefficients between the estimation sample frequency rate and the result reported in the corresponding row.
³ Number of bankrupt: number of nonbankrupt firms in the choice-based estimation sample. The bankrupt firm sample frequency rate (number of bankrupt firms/total number of sample firms) is reported in parentheses.
⁴ Percentage of firms correctly predicted.
⁵ Chi-square test comparing actual with predicted states.
⁶ Average probability of bankruptcy for the prediction sample times 100%.
⁷ Goodness-of-fit test statistic—distributed as a chi-square with six degrees of freedom. This test compares the actual portfolio frequency rate of bankrupt firms against the predicted rate across seven portfolios. The portfolios are constructed by first ranking firms on the basis of the probit index and then partitioning firms such that six bankrupt firms are in each portfolio (with the exception of the last portfolio which has five bankrupt firms).
⁸ The percentage of firms predicted to be in the same state by both the WESML and unadjusted probit assessments.
⁹ Chi-square test comparing the WESML probit predictions with the unadjusted probit predictions.
^a Significant at the .01 level.
^b Significant at the .05 level.

The correlation coefficients for the estimated parameters using unweighted probit (see table 3, panel A) are consistent with the existence of a bias: all four estimated parameters have statistically significant correlations. The positive correlation of .859 for the constant indicates that lower estimation sample frequency rates are associated with smaller (more negative) constants, and therefore have, on average, lower bankruptcy probabilities. On the other hand, only one of the estimated *WESML* parameters (rate of return, *ROA*) has a significant correlation coefficient (see table 3, panel B). The *WESML* constant has a correlation coefficient of .143 (not statistically significant at any reasonable level). Thus, the bias appears to exist in the unweighted probit results and to be eliminated in the *WESML* results.

Classification results based on the estimated bankruptcy probabilities of each estimation sample for a given technique using a .5 probability cutoff are reported in table 4. Firms with probabilities greater than or equal to (less than) .5 were classified as bankrupt (nonbankrupt).¹⁶ These results are not as useful for comparison as the prediction results reported in table 5 because of classification sample differences. The results in panel A, however, generally indicate the existence of a bias and the overclassification of bankrupt firms when using unweighted probit. The correlation coefficient between the estimation sample frequency rate and the percentage of bankrupt firms classified correctly is .949. A significant, although lower, correlation is also reported for the *WESML* estimation in panel B. This result may indicate that *WESML* does not completely eliminate the bias. However, the result may also be caused by classification sample differences, so conclusions are deferred until the prediction results are examined. Panel C of table 4 presents results comparing classifications across the two estimation techniques in two ways. The first presents the percentage of identical classifications across both techniques. The second compares the unweighted probit classifications to the *WESML* classifications using a chi-square test. Both comparisons indicate that classifications from the two techniques are more similar as the sample selection probability approaches the population probability; both have correlation coefficients measuring this trend which are negative and significant at the .01 level. Thus, the amount of bias appears to decrease as the sample frequency rate decreases.

Comparisons based on prediction tests are reported in table 5. Predictions are based on estimated bankruptcy probabilities of the prediction sample (41 bankrupt and 800 nonbankrupt firms). Bankruptcy probabilities are calculated using parameters estimated on the basis of the respective estimation technique/sample design. (A .5 cutoff probability

¹⁶ The use of a .5 probability cutoff assumes a symmetric loss function. The group error rates would change if different probability cutoffs (which imply specific asymmetric loss functions) were used; however, the comparisons across estimation samples and techniques would not be affected.

is also used to make predictions.) Comparison tests based on predictions are stronger tests than those based on classifications because the same sample is predicted for each of the alternative estimation technique/sample designs. The unweighted probit predictions are reported in panel A, the *WESML* predictions in panel B, and the comparisons in panel C of table 5.

The correlation coefficients between the estimation sample frequency rate and the percentage of firms predicted correctly based on unweighted parameters reported in panel A are all significant at the .01 level and have the expected sign. More specifically, the bankrupt firm correlation is positive (.895), indicating an overclassification bias; the nonbankrupt firm correlation is negative (−.947), indicating an underclassification bias; and the overall correlation is negative, indicating that correct prediction rates increase when parameters which are less biased are used. On the other hand, none of the correlations reported in panel B for the *WESML* predictions is significant at even the .10 level, although they are of the same sign. The chi-square tests comparing predicted with actual states are consistent with these results.

Average bankruptcy probabilities are also reported in table 5. For unweighted probit, the correlation coefficient between the average bankruptcy probability and the estimation sample frequency rate is .995 (significant at the .01 level). The range of this probability is from 4.9% for the estimation sample with a .048 frequency rate to 18.7% for the estimation sample with a .500 frequency rate. This is not the case for the *WESML*-based predictions, which have a correlation of .601 (not significant at the .10 level) and a range of 2.6 to 3.2%.

The final test examining the choice-based sample bias is a goodness-of-fit test. This test is based on a comparison of predicted bankruptcy frequency rates to actual frequency rates in portfolios of firms constructed on the basis of the probit index. Seven portfolios were constructed from the prediction sample by ranking the firms on the value of the probit index and partitioning the ranked firms such that six bankrupt firms entered each portfolio (with the exception of the seventh portfolio, which contained five bankrupt firms). The actual portfolio frequency (six for portfolios one through six and five for portfolio seven) was compared to the predicted frequency (which is equal to the sum of the estimated probabilities within each portfolio) using a goodness-of-fit test. This test (distributed as a chi-square with six degrees of freedom) indicates how closely the predicted probability distribution maps the population probability distribution. The lower the chi-square value, the better the mapping.

The null hypothesis of identical distributions is rejected (at the .01 level) four times in the six estimation sample tests based on unweighted probit and only once (at the .05 level) based on *WESML* probit. The two instances in which the null hypothesis is not rejected in the unweighted probit assessments occur in the two estimation samples which have the

lowest bankrupt firm frequency rates. This result, in conjunction with the positive and significant correlation between the unweighted probit chi-square statistics and the estimation sample frequency rates, indicates that the degree of bias decreases as the sample selection probability approaches the population probability. The comparisons reported in panel C of table 5 compare the unweighted probit predictions with the *WESML* predictions and provide equivalent conclusions to the conclusions drawn from classification comparisons reported in table 4.

The tests reported in this section demonstrate empirically the effects of using estimation techniques on choice-based samples which assume exogenous random sampling. The bias is shown to be an increasing function of the difference between the sample selection probability and the population probability. The tests also demonstrate how using *WESML* to estimate such models on choice-based samples eliminates most, if not all, of the bias.

4. Complete Data Criterion: Sample Selection Bias

Another sample selection criterion used in financial distress research is the complete data criterion. This criterion affects the estimation of a model because the likelihood of the sample on which the model is estimated is conditional on having complete data. If the probability of distress given complete data is significantly different from the probability of distress given incomplete data, the estimated model will be biased. Thus, to assume that this bias does not exist one must first assume that firms with incomplete data are a random sample of firms from the population. Incorrectly making this assumption results in asymptotically biased parameter and probability estimates (see Heckman [1979], Poirier [1980], and Maddala [1983]). In many situations one is forced to make this assumption because no alternatives exist. However, if some data are available on all firms, a model can be formulated to predict if a firm will have complete data, and the estimated probability of a firm having complete data can be directly incorporated into the estimation of the original model. A discussion of a bivariate probit approach to this problem using the model presented in equation (1) follows.

Assume that equation (4) appropriately describes the probability of bankruptcy as a probit equation where a firm is observed to file a petition for bankruptcy ($B = 1$) when B^* , an underlying response variable, exceeds zero:

$$P(B = 1) = P(B^* > 0) \quad (4a)$$

$$B^* = a_0 + a_1ROA + a_2FINL + a_3LIQ + u_B \quad (4b)$$

$$P(B^* > 0) = P(-u_B < a_0 + a_1ROA + a_2FINL + a_3LIQ) \quad (4c)$$

where u_B = a random error term and all other terms are defined as in equation (1).

Assume further that equation (5) appropriately describes the probability of a firm having complete data as a probit equation where the firm is observed to have complete data ($D = 1$) when D^* , an underlying response variable, exceeds zero:

$$P(D = 1) = P(D^* > 0) \quad (5a)$$

$$D^* = b_0 + b_1 LIST + u_D \quad (5b)$$

$$P(D^* > 0) = P(-u_D < b_0 + b_1 LIST) \quad (5c)$$

where:

$D = 1$ if complete data, 0 otherwise,

$LIST$ = number of days listed on the *CRSP Daily Return File*, and

u_D = a normally distributed error term.

When equation (4) is estimated on a sample of firms with complete data, the expected value of B^* becomes:

$$E(B^*) = E(a_0 + a_1 ROA + a_2 FINL + a_3 LIQ) + E(u_B | D^* > 0). \quad (6)$$

Thus, the estimated model (the sample model) is not the population model. The value of $E(u_B | D^* > 0)$ depends on the covariance between u_B and u_D . If u_B and u_D have a zero covariance, then $E(u_B) = E(u_B | D^* > 0) = 0$. On the other hand, if u_B and u_D have a nonzero covariance, then $E(u_B | D^* > 0) \neq 0$, and the estimated model will be biased. A negative (positive) covariance results in probability estimates which are biased downward (upward) for equation (4). Thus, obtaining estimates of equation (4) becomes more complicated if incomplete data occur in a nonrandom fashion, that is, if u_B and u_D have a nonzero covariance.

If it is assumed that u_B and u_D are distributed bivariate normal and $\sigma(u_B) = \sigma(u_D) = 1$, then:

$$E(u_B | D^* > 0) = \rho(u_B, u_D)[\phi(b_0 + b_1 LIST)/\Phi(b_0 + b_1 LIST)] \quad (7)$$

where:

$\sigma(\cdot)$ = standard deviation of $(\cdot)$,
 $\rho(u_B, u_D)$ = correlation between u_B and u_D , and
 ϕ, Φ = the density and cumulative density function for a standard normal variable, respectively.

In the case where $\rho(u_B, u_D) \neq 0$, equation (4) must be estimated using a bivariate probit assessment which incorporates the impact of missing data on estimates of bankruptcy probabilities. The appropriate specification of the likelihood function for this assessment is composed of three factors corresponding to the three subsamples: bankrupt firms with complete data, nonbankrupt firms with complete data, and firms with incomplete data.¹⁷

The sample selection bias issue is examined empirically by estimating the model presented in equation (4) using both a “simple” probit assessment (i.e., ignoring the incomplete data problem) and a bivariate probit assessment (i.e., incorporating the effects of incomplete data into the estimation). The population of firms, data sources, and data collection procedures are as described in section 2. The model is estimated on the population of firms in each year (1972 through 1978) to avoid a choice-based sample bias. The results of these analyses are reported in table 6, with the simple probit results reported in panel A and the bivariate results in panel B.

The financial distress model estimations reported in panels A and B indicate that both estimation techniques yield equivalent statistical inferences: the overall model is significant at the .01 level and two of three hypothesized relationships, as well as the constant, are statistically significant. Specifically, the results for both techniques indicate that the probability of bankruptcy is a decreasing function of return on assets (*ROA*) and an increasing function of financial leverage (*FINL*). (The liquidity coefficient [*LIQ*] has the expected negative sign in three of seven years but is never statistically significant.) These results are also consistent across years. Once again, it is unknown if the similarities between the two estimations are due to the existing relationships or to the similarity in the techniques themselves. Thus, it is unclear under what conditions bivariate probit would result in statistical inferences different from those reported using simple probit.

The complete data model estimation reported in the bivariate probit results (panel B) also indicates a significant relationship. The probability of complete data is indicated to be an increasing function of the number of days a firm is listed on the *CRSP Daily Return File (LIST)*. This relationship is observed in all years examined. The percentage of firms correctly classified by the complete data model has a range of 71.7 to 72.2%, with a range of 52.3 to 54.6% for the complete data group and 82.5 to 83.5% for the incomplete data group. The chi-square statistics comparing actual to classified (complete data) states are significant at the .01 level in all years.

The most interesting result is the estimated sample selection bias measure reported in the last row of table 6. This measure is an estimate

¹⁷ See Heckman [1979], Poirier [1980], and Maddala [1983] for a more detailed discussion of the likelihood function.

of the correlation between u_B and u_D . A significant correlation indicates the existence of a significant sample selection bias. The estimated correlations reported in table 6 range from $-.303$ to $-.719$ and are statistically significant in five of seven years. The negative correlation indicates that firms with high bankruptcy probabilities have low probabilities of having complete data. Thus, it is those firms with high bankruptcy probabilities that are excluded from estimating the model. Estimating the model on this sample without adjusting the bias results, on average, in understated bankruptcy probabilities. The classification results demonstrate this bias. Although the chi-square statistics comparing actual with classified (bankruptcy) states are statistically significant for both estimations, the percentage of bankrupt firms correctly classified using bivariate probit probability estimates is greater than or equal to the simple probit correct classifications in all years. The overall classification rates, however, are similar across estimation techniques.

The analyses conducted here demonstrate the effects of sample selection bias in financial distress research. The results, however, do not indicate different statistical inferences or different overall classification rates. Only the probability estimates appear to be affected by the bivariate probit assessment. Although no conclusive statements can be made on the basis of these results, it appears that the refinement of using a bivariate probit assessment provides few additional insights concerning statistical inferences.

5. *Summary and Concluding Comments*

This paper examines two potential biases caused by sample selection/data collection procedures used in most financial distress studies. The first bias, a choice-based sample bias, results when a researcher first observes the dependent variable and then selects a sample based on that knowledge, that is, the probability of a firm entering the sample depends on the dependent variable's attributes. The second bias, a sample selection bias, results when only observations with complete data are used to estimate the model and incomplete data observations occur nonrandomly. Both biases result in asymptotically biased parameter and probability estimates.

The choice-based sample bias is examined by comparing probit estimates for a financial distress model to estimates from an adjusted probit assessment of the model (weighted exogenous sample maximum likelihood probit) across alternative samples designed to induce increasing amounts of the bias. The results clearly demonstrate the existence of a bias for choice-based samples when unadjusted probit is used, decreases in the bias as the sample composition approaches the population composition, and the elimination of the bias using the adjustment procedure. However, the bias does not, in general, affect the statistical inferences or

TABLE 6
Comparison of Simple and Bivariate Probit Assessments of the Financial Distress Model¹

	YEAR ²						
	1972	1973	1974	1975	1976	1977	1978
<i>Panel A—Simple Probit</i>							
Coefficients ³							
CONSTANT	-3.108 ^a	-4.770 ^a	-4.631 ^a	-4.429 ^a	-4.110 ^a	-4.687 ^a	-4.188 ^a
ROA	-5.670 ^a	-4.757 ^a	-3.155 ^a	-.946 ^a	-2.971 ^a	-3.557 ^a	-2.894 ^a
FINL	1.296 ^b	4.958 ^a	4.682 ^a	4.206 ^a	4.430 ^a	4.961 ^a	4.144 ^a
LIQ	-.056	.002	.007	.011	-.270	-.122	.003
Correct Classification ⁴							
BANKRUPT	28.6	30.0	23.1	9.1	23.1	40.0	29.4
NONBANKRUPT	99.9	99.9	99.9	99.9	99.9	99.9	99.8
OVERALL	99.6	99.4	99.3	99.3	99.2	99.6	99.1
CHI-SQUARE ⁵	167.79 ^a	195.25 ^a	187.79 ^a	10.87 ^a	149.13 ^a	383.77 ^a	230.27 ^a
Log-Likelihood ⁶	-22.29 ^a	-30.74 ^a	-39.68 ^a	-35.59 ^a	-30.20 ^a	-23.56 ^a	-40.86 ^a
<i>Panel B—Bivariate Probit</i>							
EQUATION (4)—BANKRUPTCY							
Coefficients ³							
CONSTANT	-4.177 ^a	-3.171 ^b	-3.393 ^a	-2.816 ^a	-2.766 ^a	-3.841 ^a	-4.061 ^a
ROA	-3.792 ^a	-4.366 ^b	-2.615 ^a	-.846 ^a	-1.864 ^a	-2.390 ^a	-2.261 ^a
FINL	3.744 ^a	3.135 ^b	3.430 ^a	2.780 ^a	3.118 ^a	4.035 ^a	4.277 ^a
LIQ	.019	.002	.006	-.063	-.220	-.032	.002
Correct Classifications ⁴							
BANKRUPT	42.9	40.0	23.1	9.1	30.8	40.0	41.2
NONBANKRUPT	99.9	99.8	99.9	99.9	99.9	99.9	99.9
OVERALL	99.7	99.4	99.2	99.2	99.3	99.6	99.2
CHI-SQUARE ⁵	351.02 ^a	273.69 ^a	149.33 ^a	11.13 ^a	244.90 ^a	383.77 ^a	431.01 ^a
EQUATION (5)—COMPLETE DATA							
Coefficients ³							

CONSTANT.....	-1.540 ^a	-1.531 ^a	-1.511 ^a	-1.502 ^a	-1.504 ^a	-1.522 ^a	-1.504 ^a
<i>L</i> I <i>S</i> T (× 10 ²)	.036 ^a	.036 ^a	.035 ^a	.036 ^a	.034 ^a	.037 ^a	.033 ^a
Correct Classifications ⁴							
COMPLETE DATA	54.6	52.4	52.9	52.3	52.6	53.1	52.5
INCOMPLETE DATA	82.5	82.9	83.2	83.5	83.1	82.9	83.1
OVERALL	72.2	71.7	72.0	72.0	71.9	71.9	71.7
CHI-SQUARE ⁵	374.36 ^a	345.56 ^a	359.10 ^a	358.40 ^a	353.33 ^a	357.47 ^a	349.81 ^a
LOG-LIKELIHOOD ⁶	-1348.04 ^a	-1342.93 ^a	-1354.62 ^a	-1351.62 ^a	-1349.79 ^a	-1344.86 ^a	-1367.31 ^a
SAMPLE SELECTION BIAS ⁷							
$\rho(U_B, U_D)$	-.486	-.658 ^b	-.624 ^a	-.719 ^a	-.718 ^a	-.555 ^a	-.303

¹ This table presents the estimated parameter and classification results of a financial distress model (see variable definitions below) using two alternative estimation techniques: simple probit, which estimates the model conditional on complete data; and bivariate probit, which estimates both a complete data model and a financial distress model, incorporating the probability of complete data into the estimation of the financial distress model.

² Year indicates the year in which the bankrupt firms' 10K reports were received by the SEC.

³ Variable Definitions:

Financial Distress Model

ROA = net income to total assets (return on assets);

FNL = total debt to total assets (financial leverage); and

LIQ = current assets to current liabilities (liquidity)

Complete Data Model

*L*I*S*T = number of days listed on the *CRSP Daily Return File*.

⁴ The percentage of firms correctly classified using a .5 cutoff probability; firms with probabilities greater than or equal to (less than) .5 were classified as bankrupt or having complete data (nonbankrupt or having incomplete data).

⁵ Chi-square test comparing actual with classified states.

⁶ Log of the likelihood function.

⁷ The degree of sample selection bias is measured by the correlation between the residuals of the financial distress and complete data models estimated in the bivariate probit assessment.

^a Significant at the .01 level.

^b Significant at the .05 level.

the overall classification rates for the financial distress model and the samples tested.

The sample selection bias is examined by comparing probit estimates of a financial distress model conditional on complete data to estimates from a bivariate probit assessment of the model which incorporates the probability of an observation having complete data into the estimation of the financial distress model parameters. The bivariate probit model consists of two probit equations, the financial distress equation and the complete data equation. The data constraint when using bivariate probit is that all observations must have complete data for the complete data equation. The results are qualitatively similar to the choice-based sample results in that a bias is clearly shown to exist, but, in general, it does not appear to affect the statistical inferences or overall classification rates.

The potential benefit of using these approaches appears to be in estimating the sample probability distribution. The adjusted estimation techniques appear to estimate probability distributions which fit the population distribution better. This can be important if cardinal probability estimates are required (e.g., for expected value calculations) or if the effect of an independent variable on the probability estimates is of interest. However, these estimation techniques do not appear to provide different qualitative results from the results provided by techniques that assume random sampling. Only the individual group error rates are significantly affected.

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